Probabilistic Word Embeddings in Kinematic Space

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***Abstract*—In this paper, we propose a method for learning representations in the space of Gaussian-like distribution defined on a novel geometrical space called Kinematic space. The utility of non-Euclidean geometry for deep representation learning has recently been in vogue, specifically models of hyperbolic geometry such as Poincaré and Lorentz models have proven useful for learning hierarchical representations. Going beyond manifolds with constant curvature, albeit has better representation capacity might lead to unhanding of computationally tractable tools like Riemannian optimization methods. Here, we explore a pseudo-Riemannian auxiliary Lorentzian space called Kinematic space and provide a principled approach for constructing a Gaussian-like distribution, which is compatible with gradient-based learning methods, to formulate a probabilistic word embedding framework. Contrary to, mapping lexically distributed representations to a single point vector in Euclidean space, we advocate for mapping entities to density-based representations, as it provides explicit control over the uncertainty in representations. We test our framework by embedding WordNet-Noun hierarchy, a large lexical database, our experiments report strong consistent improvements in Mean Rank and Mean Average Precision (MAP) values compared to probabilistic word embedding frameworks defined on Euclidean and hyperbolic spaces. We show an average improvement of 72.68% in MAP and 82.60% in Rank compared to the hyperbolic version. Our work serves as evidence for the utility of novel geometrical spaces for learning hierarchical representations.**

I. Introduction

In this paper, we propose a probabilistic word embedding framework that learns word representations in the space of Gaussian-like distributions constructed on a novel geometrical space called Kinematic Space. Mapping entities to a density, instead of point vectors in low dimensional Euclidean spaces, has many advantages, like capturing the uncertainties in word representations and better expressing the asymmetrical relationship between words [1]. It is hypothesized that the quality of embeddings largely depends on the embedding space and how well it captures data structure. This realization has led to the resurgence of the Manifold hypothesis [2], which asserts that raw data (like pixels of images) lie on a low-dimensional manifold but with a new imperative - explicit assumption of the underlying manifold, like hyperbolic spaces [3], [4] and spherical spaces [5], [6]. Hyperbolic geometry has become the new workhorse for learning representations of data with underlying hierarchical anatomy. Authors in [7], propose to construct a Gaussian-like distribution on hyperbolic space to facilitate probabilistic inference problems with hierarchical datasets and, as an application, develop a word embedding framework that maps word representations to Gaussian-like distributions defined on hyperbolic space. Our work takes this idea further by exploring a new geometrical space called Kinematic space ($\mathcal{K}_s$). Inspired from theoretical physics [8], [9], Kinematic space acts as an intermediary space with the ability to transform information from one geometrical space to another. We start with the Poincaré upper half-plane model (H_P) of hyperbolic geometry, which is rarely considered in the literature for learning representations due to its incompatibility with Riemannian optimization tools [10], [11] and use the power of Kinematic space to transform the information from the upper half-plane model to a maximally, symmetric Lorentzian manifold called de Sitter space ($de\mathbb{S}_2$), where the appeal of Riemannian optimization tools is restored. We construct Gaussian-like distribution on the de Sitter space and formulate a probabilistic word embedding framework that maps word representations to this distribution. We experiment on an extensive lexical database called WordNet-Noun [12] and report Mean Rank (MR) and Mean Average Precision (MAP) values, which quantify the reconstruction ability of the proposed method. Our contributions can be summarized as follows -

1) We introduce a geometrical space called Kinematic Space, in the context of developing a Probabilistic Word Embedding framework,
   - that, maps lexically-distributed representations to a density instead of point vectors.
   - we do this by, constructing a Gaussian-like distribution on the Upper half-plane model of hyperbolic geometry and using Kinematic space to transform it to a computationally tractable manifold called de Sitter space.
2) We propose a series of experiments that demonstrate the effect of curvature for learning these representations.
3) We demonstrate our proposed framework by embedding WordNet-Noun hierarchy dataset and quantifying the reconstruction ability through Mean Rank and MAP values. Extensive experiments, suggest strong improvements over existing hyperbolic and Euclidean versions.

II. Related Work

In this section, we discuss few contemporaneous works that have introduced hyperbolic geometry for various deep

learning tasks. The utility of hyperbolic geometry in learning deterministic word embeddings was demonstrated by authors in [4], [3]. With the advent of Riemannian optimization methods [10], [11], a variety of works[13], [14], [15], [16], [17] have used hyperbolic geometry as an inductive bias for deep learning tasks. Authors in [18], have proposed Hyperbolic Neural Network, in which they re-derive mainstream neural network operations in the Poincaré ball model of hyperbolic geometry. Authors in [19], [20] propose a variant of Graph Neural Network architecture, where the nodes of the graphs are embedded in the hyperboloid model of hyperbolic geometry to better capture the structural attributes of the graphs for tasks like link prediction and chemical property prediction of molecules. To account for varying structure of data, authors in [21], propose a heterogeneous curvature product space instead of single constant curvature space (like hyperbolic geometry) to learn representations. Authors in [7], propose to construct a pseudo-hyperbolic Gaussian distribution on hyperbolid model and as applications develop probabilistic word embeddings and hyperbolic-version of Variational Auto Encoders (VAEs) to facilitate solving probabilistic inference problems on hyperbolic space. Results from Network Science also advocates the existence of hyperbolic geometry as the latent space for many applications, for example, authors in [22], show that the properties of complex networks such as heterogeneous degree distributions and strong clustering are reflections of the negative curvature of latent hyperbolic geometry underlying the complex networks. The utility of Kinematic Space for learning hierarchical representations has been mathematically formulated by authors in [23]. To the best of our knowledge, this is the first work that leverages Kinematic space as an embedding space for Probabilistic Word Embedding framework.

III. Preliminaries and Background

In this section, we recall few definitions of tools of differential geometry that are useful for learning representations and then we discuss the various models of hyperbolic geometry used in the literature. We then introduce and provide a mathematical formulation for Kinematic Space.

A. Review of Differential Geometry

1) **Manifold -** Let $\mathcal{M}$ be a n-dimensional topological space. The Riemannian metric g_x is defined as a collection of inner products $g_x : \mathcal{T}_x\mathcal{M} \times \mathcal{T}_x\mathcal{M} \rightarrow \mathbb{R}$ at each point $x \in \mathcal{M}$. A smooth manifold equipped with the Riemannian metric is called a Riemannian Manifold ($\mathcal{M}$, g). The metric tensor is used to define geometrical concepts like length and angles on the Riemannian manifold.
2) **Tangent Space -** The tangent space $\mathcal{T}_x\mathcal{M}$ of a n-dimensional manifold $\mathcal{M}$ is the set of all possible direction curves passing through the point $x \in \mathcal{M}$. The elements of $\mathcal{T}_x\mathcal{M}$ are the tangent vectors.
3) **Geodesics -** A curve $\gamma \in [a, b]$ is a generalization of straight lines to constant speed curves called Geodesics, giving the shortest path between points $\{x, y\} \in \mathcal{M}$. The length of the curve is calculated as $\mathsf{L}(\gamma) = \int_a^b ||\bar{\gamma}(t)||dt$. Global distances induced on $\mathcal{M}$ can be defined as $d_{\mathcal{M}}(x, y) = \inf_\gamma\{\mathsf{L}(\gamma)\}$.
4) **Parallel Transport -** The Parallel transport $PT_{x\rightarrow y}(.)$ is a generalization of translation to non-Euclidean geometry. For a given smooth manifold $\mathcal{M}$, the parallel transport $PT_{x\rightarrow y}(.)$ maps a vector $\mathbf{v} \in \mathcal{T}_x\mathcal{M}$ to $PT_{x\rightarrow y}(\mathbf{v}) \in \mathcal{T}_y\mathcal{M}$. The Riemannian metric tensor is preserved.
5) **Exponential and Log Map -** Let x be a point on the manifold $\mathcal{M}$ and v be a tangent vector of the tangent space $\mathcal{T}_x\mathcal{M}$. The exponential map $\exp_x : \mathcal{T}_x\mathcal{M} \rightarrow \mathcal{M}$ allows for mapping vectors $v \in \mathcal{T}_x\mathcal{M}$ to some point $q \in \mathcal{M}$ by using a geodesic $\gamma_{(x,v)}(t)$. The logarithmic map is the inverse of exponential map defined as $\log_x = \exp_x^{-1} : \mathcal{M} \rightarrow \mathcal{T}_x\mathcal{M}$

A combination of these geometrical concepts help in constructing the said Gaussian-like distribution in our prescribed geometry.

B. Models of Hyperbolic Geometry

In this section, we review some of the most commonly used models of hyperbolic geometry for deep representation learning tasks.

1) **Poincaré ball model -** is a Riemannian manifold $(\mathcal{B}, g_B)$, where $\mathcal{B} = \{x \in \mathbb{R}^n : ||x|| < 1\}$ is an open unit ball. The distance between points $x, y \in \mathcal{B}$ is given by,

$$d_{\mathcal{B}}(x, y) = \operatorname{arcosh}\left(1 + 2\frac{||x - y||^2}{(1 - ||x||^2)(1 - ||y||^2)}\right) \tag{1}$$

2) **Lorentz Model -** aka hyperboloid model, is a Riemannian manifold $(\mathcal{L}, g_L)$, where $\mathcal{L} = \{\mathbf{x} \in \mathbb{R}^{n+1} : \langle \mathbf{x}, \mathbf{x}\rangle_{\mathcal{L}} = -1, x_o > 0\}$ and the Minkowski inner product defined by,

$$\langle x, y\rangle_{\mathcal{L}} := -x_0y_0 + x_1y_1 + ... + x_ny_n \tag{2}$$

The induced distance function is given by,

$$d_{\mathcal{L}}(\mathbf{x}, \mathbf{y}) = \operatorname{arcosh}(-\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}}) \tag{3}$$

3) **Poincaré Upper Half-Plane Model:** is a Riemannian manifold $\mathbb{H}_{UP} = \{(\mathbf{x}, \mathbf{y} \in \mathbb{R}), \mathbf{y} > 0\}$, its distance metric is given by,

$$d_{\mathbb{H}_{UP}}((x_1, y_1), (x_2, y_2)) = \operatorname{arcosh}\left(1 + \frac{(x_2 - x_1)^2 + (y_2 - y_1)^2}{2y_1y_2}\right) \tag{4}$$

The Lorentz model of hyperbolic geometry is the most commonly used geometrical space in literature [13], [3] as it is numerically stable. None of the literature, talks about the utility of the upper-half plane model since it is not computationally tractable, owing to the presence of fraction in the geodesic distance (Equation 4). In the next section, we

introduce Kinematic space and how it enables for transforming information from one space to other.

IV. Why Kinematic Space?

Many recent works [13], [18], [14], [15], [24], by default employ hyperbolic space as the embedding space for learning hierarchical representations. However, many real-world graphs also exhibit properties such as *Causality*, (for example, citation networks), which is not captured by hyperbolic geometry. As authors in [3] describe, tasks like - inferring concept hierarchies in large taxonomies of data involves defining a *partial order set* (Posets) over entities, and the chosen embedding space should be capable of preserving this order. These partial order sets are also called as *Causal Sets* in theoretical physics and are directly related to Lorentzian Manifolds [25]. Thus, the topology of structural datasets with hierarchical anatomy and/or expressing causal relationships are more naturally described by Lorentzian manifolds such as de Sitter ($d\mathbb{S}_2$) than hyperbolic spaces. We substantiate the motivation for proposing Kinematic space, by embedding[1] Arxiv-HEP Citation dataset [26] and SCOTUS dataset [27] in both the Poincaré Disk ($\mathbb{D}_P$) and de Sitter space $d\mathbb{S}_2$ by using the Lorentzian Multi-dimensional scaling algorithm proposed by authors in [28].

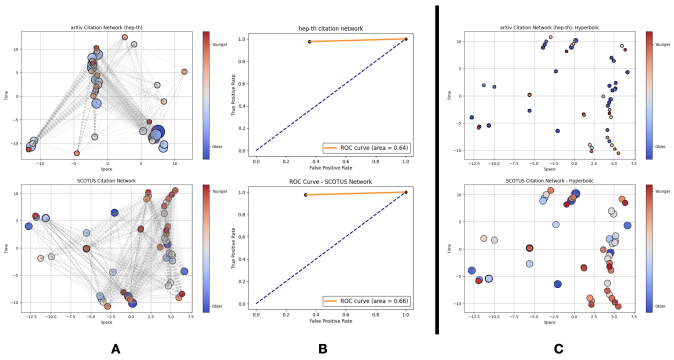

Figure 1. We embed two citation networks [27], [26] using the MDS algorithm proposed by authors in [28]. Since the graphs exhibit both causality and hierarchical nature, the proposed de Sitter space better embeds the graph (Figure A).The ROC curve (Figure B) measures the accuracy of embedding the graph in the proposed geometrical space. Embedding in the Poincaré Disk fails (Figure C). Note, the distribution of graph w.r.t the time coordinate.

A. Kinematic Space

In this section, we provide the mathematical background for Kinematic space ($\mathcal{K}_s$). Kinematic space, is an intermediary, auxiliary geometrical space primarily introduced in the context of theoretical physics for quantum gravity [8], [9]. The Crofton's formula [29] allows us to calculate the length of any arbitrary curve $\mathcal{C}$ by describing its length as the volume of tangents intersecting the curve. For a given equation of straight line $x\cos\theta + y\sin\theta - l = 0$, where θ is the polar angle and l is the distance of straight line from the origin as shown in Figure 2 A. The Crofton's formula is given by,

[1]We use code https://github.com/drewwilimitis/hyperbolic-learning

$$\text{Length} = \frac{1}{4}\int_0^{2\pi} d\theta \int_{-\infty}^{+\infty} \eta(\theta, l)dl \tag{5}$$

where $\eta(\theta, l)$ gives the intersection number of the tangents and the curve. This space of oriented geodesics panning $\theta \in [0, 2\pi]$ and $l \in (-\infty, \infty)$ is called the *Kinematic space of the Euclidean plane*. For some curve located on the hyperbolic plane ($\mathbb{H}^n$), the straight lines are now replaced by geodesics given by equation $\tanh\rho\cos(\hat{\theta} - \theta) = \cos\alpha$ where, α is the opening angle of the geodesic and θ is the angular coordinate of the center of the geodesic. The Kinematic space of the hyperbolic plane is now the space of geodesics panned by $\theta \in [0, 2\pi]$ and $\alpha \in [0, \pi]$. This is represented in Figure 3, where a geodesic γ is drawn on the Poincaré upper half-plane model (H_P), its stereographic projection is shown on the Poincaré disk (P_d). Thus, the Kinematic space is an auxiliary Lorentzian space of oriented geodesics that acts as an intermediary for translating information from the space of length of curves to a space where the length of the curves can be expressed as volume of intersecting lines.

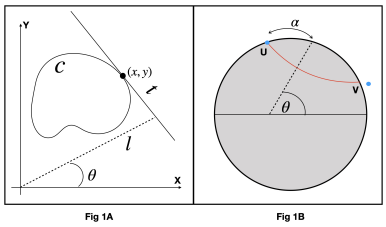

Figure 2. The Crofton's formula can be used to estimate the length of a closed curve $\mathcal{C}$. Parametrized by (θ, l), we can express the length of the curve as a volume of straight lines intersecting it (Figure 1A). Straight lines are replaced by geodesics γ on hyperbolic plane and can be parametrized by (θ, α) as shown in (Figure 1B).

B. de Sitter Space as the Kinematic Space

In this section, we provide a mathematical formalism for translating information from the Upper half plane model to de Sitter space. The de Sitter space is a maximally symmetric, Lorentzian manifold with constant positive curvature. By using the formulation outlined in Section IV-A we show that the Kinematic space of the Upper-half plane model is the de Sitter space, and use it for learning representations. The proof is available in the supplementary section, here we directly use the result.

Proposition 1. The Kinematic space ($\mathcal{K}_s$) of the Upper half-plane model ($\mathbb{H}_{UP}$) is the $(d+1)$ dimensional de Sitter space ($de\mathbb{S}_2$) visualized as a single sheeted hyperboloid in the Minkowski space $\mathbb{M}^{2+1}$ and there is an identification between the geodesics $\gamma \in \mathbb{H}_{UP}$ and points in $\mathcal{K}_s$.

Proposition 2. Let $de\mathbb{S}_2$ be the $(d+1)$ dimensional de Sitter space in the $(d+2)$ dimensional Minkowski space $\mathbb{M}$ visualized as a single sheeted hyperboloid with pseudo-radius λ given by $-z_0^2 + z_1^2 + z_3^2 + ... + z_n^2 = \lambda^2$. The metric is induced

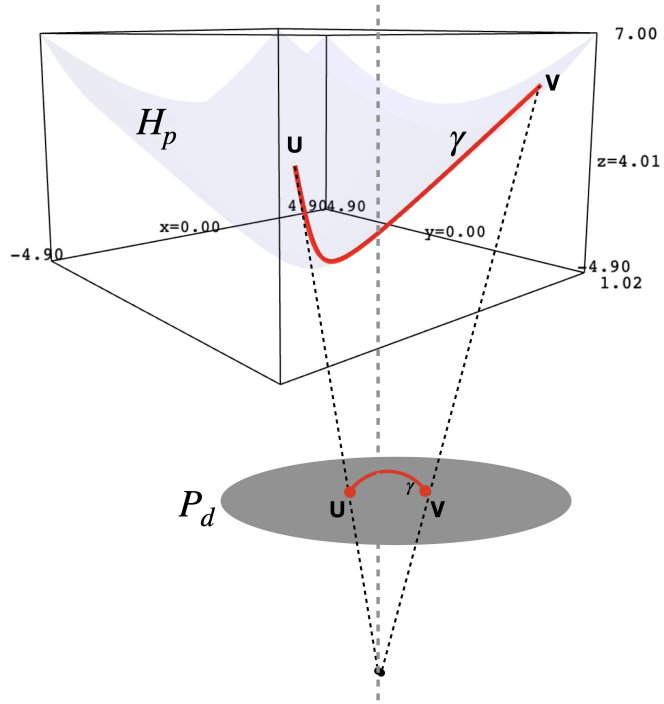

Figure 3. A geodesic γ drawn on the upper half-plane model can be stereographically projected on to the Poincaré disk, parameterized by the opening angle α and the angular coordinate θ. This space of oriented geodesics is the Kinematic Space of the Hyperbolic Plane.

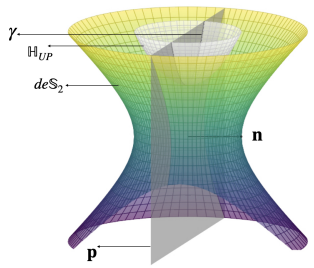

Figure 4. The Kinematic space $\mathcal{K}_s$ of the Upper half-plane model $\mathbb{H}_{UP}$ is the de Sitter space $de\mathbb{S}_2$ visualized as a single-sheeted hyperboloid with its distance metric induced by the ambient Minkowski space.

by the embedding Minkowski space [30], where $\lambda^2 = \frac{1}{K}$ and the distance function is given by,

$$d_{de\mathbb{S}}(\mathbf{x}, \mathbf{y}) = \lambda \operatorname{arcosh}\left(\frac{-\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}}}{\lambda^2}\right) \tag{6}$$

The exponential map $\exp_x : \mathcal{T}_x de\mathbb{S}_2 \rightarrow de\mathbb{S}_2$ and the logarithmic map $\log_x : \mathcal{T}_x de\mathbb{S}_2 \rightarrow de\mathbb{S}_2$ is given by,

$$\exp_x(\mathbf{v}) = \cosh(\sqrt{K}||\mathbf{v}||_{\mathcal{L}})\mathbf{x} + \mathbf{v}\frac{\sinh(\sqrt{K}||\mathbf{v}||_{\mathcal{L}})}{\sqrt{K}||\mathbf{v}||_{\mathcal{L}}} \tag{7}$$

$$\log_x(\mathbf{y}) = \frac{\operatorname{arcosh}(-K\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}})}{\sinh(\operatorname{arcosh}(-K\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}}))}(\mathbf{y} - K\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}}\mathbf{x}) \tag{8}$$

where, $||\mathbf{v}||_{\mathcal{L}} = \sqrt{\langle v, v\rangle_{\mathcal{L}}}$ is the norm of Minkowski inner product (Equation 2) and K denotes the curvature of the de Sitter space. The curvature provides insights of how the shape of the geometrical space manifests. For $K = \{0, +, -\}$, depending on the signature, the space can be either flat, positively curved or negatively curved respectively. Thus, Kinematic Space acts as an intermediary transformation space that converts information from one geometrical space to other. In our case, we use it to convert information about geodesics and induced distance function defined on the upper-half plane model to the de Sitter space.

V. Constructing Gaussian-like Distribution

Our formulation of the Gaussian-like distribution $\mathcal{G}_{de\mathbb{S}_2}(\mu, \Sigma)$, where $\mu \in de\mathbb{S}_2$, is inspired from [31]. The steps involved in construction of Gaussian-like distribution on de Sitter space can be summarized as follows:

1) Sampling a vector $\mathbf{v}$ from the Gaussian distribution $\mathcal{N}(0, \Sigma)$ defined over $\mathbb{R}^n$.
2) Parallel transporting $\mathbf{v}$ from the tangent space $\mathbf{o}$ to the tangent space of new point $\mathbf{u}$ to obtain $\mathbf{w}$ by using the formula,

$$PT_{\mathbf{o}\rightarrow\mathbf{u}}(\mathbf{v}) = \mathbf{v} - \frac{K\langle y, u\rangle_{\mathcal{L}}}{1 + K\langle o, u\rangle_{\mathcal{L}}}(o + u) \tag{9}$$

3) Map the point $\mathbf{w}$ to the manifold using the exponential map at $\mathbf{u}$ given by the Equation 7.

The probability density function can be obtained by using the formula [24],

$$\log p(\mathbf{z}) = \log p(\mathbf{v}) - (n-1)\log\frac{\sinh(||\mathbf{w}||_{\mathcal{L}})}{||\mathbf{w}||_{\mathcal{L}}} \tag{10}$$

VI. Probabilistic Word Embeddings

In this section, we discuss how we can map the lexically distributed representations to the Gaussian-like distribution constructed on the de Sitter space. To train a probabilistic word embedding model, we denote (s $\sim$ t) to define some relation between words s and t. We use the objective function defined by authors in [1],

$$\mathcal{L}(\theta) = \mathbb{E}_{(s\sim t, s\nsim t')}[\max(0, m + E(s,t) - E(s,t'))] \tag{11}$$

where, $\mathbb{E}_{(s\sim t)}$ denotes the similarity between words s and t quantified by their respective distributions $\{q_s, q_t \in \mathcal{G}_{de\mathbb{S}_2}(\mu, \Sigma)\}$. We minimize the KL divergence $D_{KL}(q_s||q_t)$ between the distributions. We embed WordNet:Noun hierarchy dataset and evaluate the reconstruction performance.

A. Dataset

WordNet - is a large lexical database of English [12] consisting of nouns, adverbs, verbs grouped as a set of 'synsets' or synonyms having a hyperonymy relation. We use the WordNet-Noun hierarchy.

B. Training Details

- Number of Iterations = 50000.
- Burn-in Step = 25000.
- Batch size = 1000 for WordNet - Noun dataset.
- Curvature $K = \{2, 3, 5\}$.
- Learning rate = 0.5.

All the implementation was done on Tesla K80 GPU provided by the Google Colab environment [2]. We adopt the code [3] base of [7]. All the experiments were averaged over 2 trials.

VII. Results and Discussions

The evaluation protocol involves measuring the Rank and Mean Average Precision (MAP) as our evaluation metrics. The objective is to achieve a lower Rank and a higher MAP value. We perform extensive experiments to substantiate the efficacy of our framework. In Table I, we compare our work with the Euclidean version of density based word representation framework proposed by authors in [1] and the hyperbolic version proposed by authors in [7] on the WordNet-Noun dataset. We vary the curvature value $K = \{2, 3, 5\}$ and keep it constant while training. From the table, its evident that our proposed framework outperforms both [1] and [7] by a large margin. Our method reports an average improvement of 72.68% in MAP and 82.60% in Rank compared to the hyperbolic version. We compare our proposed work with the Poincaré embeddings[4], which is a deterministic word embedding framework that learns representations in the Poincaré ball model but does not use density based representations on the WordNet -Noun hierarchy dataset. The results are summarized in Table II. Our proposed method outperforms Poincaré embeddings for all dimensions.

A. Effect of Curvature and δ-hyperbolicity

Learning representations in Kinematic Space is advantageous as it allows for adjusting the curvature explicitly. The curvature values from the range $K = \{2, 3, 5\}$ are selected a priori and kept constant while training. The motivation for changing the curvature might not be directly discernible. However, we hypothesize that, the performance improvement can be attributed to a property called δ-hyperbolicity [32]. Gromov's δ-hyperbolicity measures the Tree-like property of the graphs, which effectively helps decide if a given graph data is suitable for embedding in hyperbolic geometry. The lower the δ-hyperbolicity, the more hyperbolic the graph is ($\delta = 0$ for Trees) and more suitable hyperbolic geometry is, as an embedding space. Formally, the hyperbolicity $\delta(u, v, x, w)$ of a quadruple is defined as the half the difference between the biggest two of the following sums, that is, $d(u, v) + d(w, x) \geq d(u, x) + d(w, v) \geq d(u, w) + d(v, x)$, and $\delta(u, v, x, w) = (d(u, v) + d(w, x) - d(u, x) - d(w, v))/2$. To define the hyperbolicity of a graph $G(V, E)$, we can view the graph as a metric space (V, d_G), with d_G measuring the (geodesic) distance between given vertices $\{u, v\}$ and then applying the 4 point condition described. In practice, we calculate the δ_{avg} as it is more robust to random perturbations [33]. Thus, by varying the curvature, we are essentially making the space more *hyperbolic*, thus boosting the performance. In Figure 5, we have calculated the δ_{avg} for a subset of WordNet-Noun dataset, we can see that $\delta_{avg} \approx 0$, implying the dataset is best embedded in curved spaces such as hyperbolic geometry. Empirically, we found that for lower dimensions $n = \{5, 10\}$, the Mean Rank and MAP value achieved better results as the K value was increased consistently. For higher dimensions, $n = \{20, 50\}$, we found that lower K value resulted in better performance, and a further increase in the K value resulted in a decrease in the performance. This is summarized in Table III). Our proposed work's current limitation is that the optimal value of curvature for learning representations is selected through experiments. A possible solution would be to learn the curvature as a parameter while training. This would serve as an ingress for our future works.

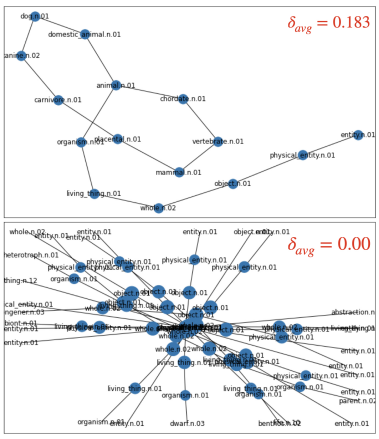

Figure 5. The $\delta_{avg} \approx 0$ implies the dataset has a hierarchical structure which is best embedded in hyperbolic space.

VIII. Conclusion

In this paper, we have proposed a density-based word embedding framework, which uses Gaussian-like distribution constructed on a novel geometrical space called Kinematic Space. The utility of Kinematic Space lies in helping transform information from one geometrical space to other, in our case we have transformed information such as geodesic distance and log/exponential map from Upper-half plane model to de Sitter space to construct the Gaussian-like distribution. We advocate for mapping lexically-distributed representations to a density, instead of mapping it to a single point vector in Euclidean space, since it has many advantages as outlined in our paper. Through extensive experiments on various standard datasets, we have shown consistent improvements over existing methods.

References

[1] L. Vilnis and A. McCallum, "Word representations via gaussian embedding," in *International Conference on Learning Representations*, 2015.

[2] https://colab.research.google.com

[3] https://github.com/pfnet-research/hyperbolic_wrapped_distribution

Table I

WE COMPARE OUR PROBABILISTIC WORD EMBEDDING FRAMEWORK WITH [1] AND [7] WHICH ALSO MAPS WORD REPRESENTATIONS TO A DENSITY ON THE WORDNET-NOUN HIERARCHY DATASET. OUR PROPOSED FRAMEWORK OUTPERFORMS BOTH THE METHODS.

	Euclid		Hyperbolic		**Ours**	
Dimension	Rank	MAP	Rank	MAP	Rank	MAP
5	70.15 $\pm$ 3.76	0.15$\pm$ 0.01	90.81 $\pm$ 8.01	0.20 $\pm$ 0.01	**4.23** $\pm$ **2.98**	**0.53** $\pm$ **0.13**
10	24.06 $\pm$ 8.85	0.43 $\pm$ 0.02	15.67 $\pm$ 4.78	0.53 $\pm$ 0.07	**1.43** $\pm$ **0.01**	**0.86** $\pm$ **0.12**
20	13.63 $\pm$ 1.69	0.65 $\pm$ 0.04	8.27 $\pm$ 2.59	0.71 $\pm$ 0.06	**2.05** $\pm$ **1.33**	**0.94** $\pm$ **0.06**
50	6.43 $\pm$ 2.17	0.75 $\pm$ 0.05	4.84 $\pm$ 0.95	0.74 $\pm$ 0.01	**1.50** $\pm$ **0.23**	**0.97** $\pm$ **0.00**

Table II

WE COMPARE OUR PROPOSED METHOD AND THE HYPERBOLIC VERSION [7] WITH THE DETERMINISTIC EMBEDDINGS FRAMEWORK PROPOSED BY AUTHORS IN [4] ON THE WORDNET-NOUN DATASET.

	Poincaré [4]		Hyperbolic		**Ours**	
Dimension	Rank	MAP	Rank	MAP	Rank	MAP
5	4.9 $\pm$ 0.00	**0.823** $\pm$ **0.00**	90.81 $\pm$ 8.01	0.20 $\pm$ 0.01	**4.23** $\pm$ **2.98**	0.53 $\pm$ 0.13
10	4.02 $\pm$ 0.00	0.851$\pm$0.00	15.67 $\pm$ 4.78	0.53 $\pm$ 0.07	**1.43** $\pm$ **0.01**	**0.86** $\pm$ **0.12**
20	3.84 $\pm$ 0.00	0.855 $\pm$ 0.00	8.27 $\pm$ 2.59	0.71 $\pm$ 0.06	**2.05** $\pm$ **1.33**	**0.94** $\pm$ **0.06**
50	3.98 $\pm$ 0.00	0.86 $\pm$ 0.00	4.84 $\pm$ 0.95	0.74 $\pm$ 0.01	**1.50** $\pm$ **0.23**	**0.97** $\pm$ **0.00**

Table III

LEARNING PROBABILISTIC WORD EMBEDDING IN KINEMATIC SPACE ALLOWS FOR CONTROLLING THE CURVATURE EXPLICITLY. CHANGING THE CURVATURE VALUE HAS A DIRECT INFLUENCE ON THE MEAN RANK AND MAP VALUE. EMPIRICALLY, WE FOUND THAT, FOR THE WORDNET - NOUN DATASET, FOR LOWER DIMENSIONS (N = 5, 10), INCREASING THE K VALUE (AND THUS DECREASING THE CURVATURE) RESULTED IN DRAMATIC INCREASE IN THE PERFORMANCE, WHILE FOR HIGHER DIMENSIONS, SMALLER K VALUES SUFFICE.

Curvature	K=2		K=3		K=5	
Dimension	Rank	MAP	Rank	MAP	Rank	MAP
5	24.77$\pm$ 18.69	0.24 $\pm$ 0.12	4.33$\pm$ 2.51	0.53$\pm$ 0.01	4.22$\pm$2.98	0.53$\pm$ 0.13
10	2.22$\pm$ 0.10	0.79$\pm$ 0.04	1.66$\pm$ 0.62	0.85$\pm$ 0.17	1.44$\pm$ 0.00	0.86 $\pm$ 0.12
20	2.05$\pm$ 1.33	0.94 $\pm$ 0.06	1.61$\pm$ 0.23	0.80$\pm$ 0.21	13.00$\pm$ 15.39	0.60$\pm$ 0.33
50	1.50$\pm$ 0.23	0.96$\pm$ 0.01	3.00 $\pm$ 0.31	0.78 $\pm$ 0.11	0.58 $\pm$ 0.29	8.94 $\pm$ 10.92

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